

Liquidity and Jumps: An Analysis on Indian Stock Market

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Abstract

Our paper is the first attempt in literature to understand the relationship of liquidity variables with the jump movements based on emerging market stocks. We use 15-minute return data from ten of the largest Indian banking stocks and implement an event study method to examine the behaviour of liquidity variables around the jump times. We find notable variations in the liquidity measures around the jump occurrences in the stocks. Liquidity characterized by market spread, trading quantity and immediacy do improve around jumps. The results indicate that the demand of immediacy of traders may cause jumps in stock returns. The Mann-Whitney test results also confirm the significant changes in the liquidity variable during the jump intervals from the non-jump intervals. Our probit and logit estimations show that the liquidity variables have more explanatory power in determining the probability of negative jump occurrences than that of positive ones. We do not observe substantial changes in the results by diving our sample into pre-crisis, crisis and post crisis periods indicating that the effects of liquidity variables are general on jump occurrences. Finally several liquidity variables are shown to contribute to price discovery though the post jump price discovery process does not seem strongly related with this liquidity variables.

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1 Introduction

Jumps, described as abrupt changes in return, are important in investors' risk management and hedging decisions. Asset prices experience infrequent jumps in addition to continuous movements. It is believed that in general the arrival of news in the market causes jumps. However, empirical studies show that a large portion of jumps can not be attributed to news arrivals. There must be other factors that are associated with jump occurrences and the importance of jumps warrants examining those factors. Not many studies have been conducted so far, in the direction of unearthing jump-related non-news variables. Our paper is an effort to bridge the gap in the existing literature by examining the relationship of liquidity variables with the jump movements of equity securities. Using ten of the largest Indian banking stocks we explore the relationship between liquidity variables and the jump returns of these stocks.

The relationship between liquidity and stock return is well studied in the literature. The effect of liquidity on the expected return may arise from the investor's recognition that they will face transaction costs when selling their stocks in the future. This will prompt discounting of stocks with higher transaction costs or capital restrictions (Batten and Vo, 2014). Amihud and Mendelson (1986) show that in equilibrium, investors with longer investment horizons will hold less liquid assets. As a result of this horizon clientele, the observed asset returns are an increasing and concave function of the transaction costs. Jumps, as a state of returns that is statistically different from other returns given the volatility of a given time window, may result from extreme changes in liquidity conditions.

Some authors argue that liquidity affects stock returns through its positive effect on corporate governance and firm performance (e.g., Karolyi, 2012). Batten and Vo (2014) discuss a number of different ways in which more liquidity may affect corporate governance, such as, by making it easier for non-block-holders to intervene and become block-holders as shown in Maug (1998), by facilitating the information of a toehold stake as shown in Kyle and Vila (1991); by reducing managerial opportunism such as in Edmans (2009), and by stimulating trade and thereby improving investment decisions by informed investors such as in Subrahmanyam and Titman (2001). The

liquidity-return relationship prompts us to anticipate that the jumps that can not be explained by news arrival may be related to rapid change in market liquidity, as argued by Lahaye et al., 2011. Boudt and Petitjean (2014) find that jumps are driven by the variations in the demand for immediacy as such jumps occur due to market inability to absorb new orders without moving the price significantly up or down.

Liquidity, by its nature, is a difficult concept to define and hence difficult to estimate (Lesmond, 2005). Kyle (1985) notes that the liquidity concept is “slippery” and “elusive”, and he attributes this to a number of transactional properties of markets encompassed in liquidity, such as tightness, depth, and resiliency. Following this, Lesmond (2005) span the definition of empirical liquidity into direct trading costs (tightness) measured by the bid-ask quotes (quoted or effective), to indirect trading costs (depth and resiliency), measured by price impact. A wide variety of liquidity measures have been used in the literature reflecting its multiple dimensions. For example Liu (2006) defines liquidity as the ability to trade large quantities quickly at low cost and with little price impact. Four dimensions, namely trading quantity (depth), trading speed (immediacy), trading cost (tightness) and price impact (resilience) emerge from this definition. As our motive is to pin down the causes of the jump, we map the different dimensions of liquidity that can be observed in the limit order book, and investigate whether there is any systematic pattern prior to or at the time of a jump.

An important measure of liquidity is trading volume or turnover. Amihud and Mendelson (1986) show that turnover is negatively related to illiquidity costs. Shahzad et al. (2014) explain the relation between volatility and volume by arguing that it is impossible for prices to vary in absence of any trading activity. Individual investors behave as noise or liquidity traders (Barber and Odean (1999)). When volume and market depth surge due to increased market activities of uninformed or noise traders, then efficiency would decrease by increasing the volatility in incorporating information. The occurrence of a jump increases uncertainty around fundamental values and it is likely to have a positive impact on future volatility. Thus, there is a practical interest in identifying the jump component of volatility and understanding how market activities contribute to jumps in volatility. Earlier studies have been mainly conducted on the US markets, with the

presence of specialists or dealers. However, whether the results for the US market can be replicated in other markets is an open question given that the majority of them are now electronic limit orders (Jain, 2005).

The interaction among public announcement, volume and stock price has been discussed in Kim and Verrecchia (1991) and Atiase and Bamber (1994) which can be summarised as follows: Each investor achieves the optimal portfolio prior to a public disclosure based on existing public and private information in the pre-disclosure period. The announcement of public information forces investors to revise their beliefs inducing both price changes and trading volume. After the announcement, investors with more precise private pre-disclosure information will on average make smaller belief revisions than less informed investors with less precise private information. Thus pre-disclosure information asymmetry result in differential belief revisions, in turn, induces trading volume. Thus it can be shown that trading volume reaction is proportional to both the absolute price change and level of pre-disclosure information asymmetry. When the pre-disclosure information asymmetry is abnormally high trading volume reaction will be high enough to induce jumps in the stock price.

An important measure of liquidity is the bid-ask spread or depths at quoted bid and ask prices (Amihud et al., 2006). Amihud and Mendelson (1986) propose that illiquidity can be measured by the cost of immediate execution which can be explained by a trade-off faced by an investor willing to transact. The investor can either choose waiting to transact when the market price matches with his desired price or choose to execute immediately at the current bid or ask price. The quoted ask and bid prices includes a premium or concession for immediate execution of the trade. Thus Amihud and Mendelson (1986) shows that a natural measure of illiquidity is the spread between the bid and ask prices which is the sum of the buying premium and selling concession. Indeed the relative quoted spread on stocks has been found to be negatively correlated with liquidity characteristics such as the trading volume and stock price continuity. The authors predict that the higher-spread assets yield higher expected return. in other words the bid-ask spread, which is often called the quoted spread, is a transaction cost to the trader for immediacy (Copeland and

Galai, 1983).

Another spread measure, the effective spread is considered to be one of the best proxies for stock liquidity. The effective spread defined by Fang et al. (2009) as the difference between the execution price and the midpoint of the prevailing bid-ask quote (the effective spread). Like the quoted spread measure, the effective spread is standardized to adjust for the stock price level converting it to a relative effective spread measure. Liquidity proxies calculated using low frequency stock returns are frequently compared to benchmark liquidity measures calculated using high frequency data to judge their effectiveness as a liquidity proxy. Statman et al. (2006) show that high returns lead to additional trading activity, hence reverse causality is a potential concern for liquidity proxies that rely on trading activity as a measurement input. Fang et al. (2009) argue that the effective spread measure is less subject to this concern than the other measures of liquidity.

In addition to liquidity conditions, liquidity shocks may play an important role in asset (fixed income assets in their paper) price jumps as suggested by Jiang et al. (2011) . In their analysis, liquidity shocks carries a broad meaning, and it could arise due to pure trading imbalance or order withdrawal. As discussed in Fleming and Piazzasi (2006) dealers tend to withdraw their orders to avoid being picked off in the upcoming information event. Jiang et al. (2011) define shock in spread as the difference between overall spread in an interval and the mean of overall spread over the previous five intervals scaled by its standard deviation.

Trading imbalance is another variable that provides a link between trading activity and returns Chordia et al. (2002). Order imbalances which show imbalance between buy orders less sell orders or excess buy or sell orders, reduce liquidity. Chordia et al. (2002) and Chordia and Subrahmanyam (2004) argue that market-wide returns are strongly affected by contemporaneous and lagged order imbalances. They show that prices and liquidity should be more strongly affected by more extreme order imbalances, regardless of volume, for two reasons. First order imbalances sometimes signal private information, thus reducing liquidity at least temporarily. Second, large order imbalances expose the inventory problem faced by the market maker, who in turn may respond by changing bid-ask spreads by changing price quotations. Hence, order imbalances should be important influences

on stock returns and liquidity, conceivably even more important than volume.

To the best of our knowledge, the only three papers that study the liquidity dynamics around jumps are Jiang et al. (2011), Boudt and Petitjean (2014) and Frömmel et al. (2013). Among these papers only Boudt and Petitjean (2014) examined jumps in stock returns while Jiang et al. (2011) used bond returns and Frömmel et al. (2013) use exchange rate returns to find the effect of liquidity. All of these papers have been conducted on the U.S market which is driven by the market dealer or market makers. Most of the theories discussed in the literature on market activities is based on how the market maker reacts to a change in market activity. Hence, the findings may be different in a market like India which is order driven rather than quote driven. We focus on stocks of the National Stock Exchange in India (NSE) where trading takes place through an open electronic limit order book, in which order matching is done by the trading computer. There are no market makers or specialists and the entire process is order driven, which means that market orders placed by investors are automatically matched with the best limit orders.

The argument in favour of choosing India for this study follows the argument made by Bekaert et al. (2007). The growing body of research on liquidity primarily focuses on the U.S., arguably the most liquid market in the world. In contrast, our research focuses on an emerging market where liquidity effects may be particularly strong. Emerging markets are known to be less liquid, hence higher liquidity premia should be found in these markets. Thus the focus on emerging markets should yield powerful evidence of liquidity impacts on any variable like jumps. In addition, the vastness of the U.S. market allows a very diversified ownership structure, combining long horizon investors (less subject to liquidity risk) with short-term investors. Hence, the pricing of liquidity is mitigated by the clientele effects in portfolio choice in this market. Such diversity in securities and ownership is lacking in emerging markets, potentially strengthening liquidity effects.

We have chosen ten banking stocks with the largest market capitalization. In countries like India banking stocks are popular, and are among the stocks that are most followed by investors and analysed by experts. Thus, we expect that there will be considerable amount of variations in liquidity variables with the changing market conditions, which enables us to capture the dynamics

between liquidity and jumps.

We use the jump test developed by Lee and Mykland (2008). This method has the advantage of identifying jumps in individual return observation. We observe jumps in 1.58% to 1.73% of return observations in the ten banking stocks. In all cases the number of positive jumps is higher than the number of negative jumps. We implement a non-parametric method to analyse the link between intra-day liquidity dynamics and properly identified intra-day jumps regardless of the arrival of information. To complement the non-parametric event study we apply a parametric analysis to assess the contribution of liquidity shocks in both the occurrence of jumps and price discovery process. We find that liquidity measures are largely affected by jumps in the stocks. If we characterize liquidity by the market spread, jumps do improve market liquidity conditions which is in contrast with the findings of Boudt and Petitjean (2014). If trading quantity and immediacy are considered, we observe again an improvement in the market liquidity. The surge in volume can be translated as a sharp increase in investors' order for immediate execution around jumps. Among the two drivers of volume surges, the rise in average trade size and the rise in number of trades, our analysis reveals that the former one is more prominent than the latter one. At the same time, market depth is deepened at the time of jumps. We show that negative jumps are associated with higher bid and ask depths, indicating the existences of thicker buying and selling pressures. Thus it is the heightened demand of immediacy from the market participants of both sides, rather than the traders' withdrawal of liquidity that causes jumps. The Mann-Whitney test results also confirm the significant changes in the liquidity variable during the jump intervals from non-jump intervals.

We examine the contemporaneous and lagged effect of liquidity factors on the probability of jump occurrences, and find the prevalence of both effects in different degrees for different jump measures. However, the explanatory power of contemporaneous liquidity variables are much higher than the lagged variables for all jumps (19.17% vs 1.54% in the probit regression). The liquidity variables have more explanatory power in determining the probability of negative than that of positive jumps (35% vs 11.53% in the contemporaneous probit regression and 2.18% vs. 1.29%

in the lagged probit regression). Negative jumps are more associated with change in liquidity condition than the positive jumps. Among the liquidity variables, volume, quoted spreads, effective spread shock and order imbalance are key drivers of the occurrences of positive jumps. For negative jump occurrences the effective spread and effective spread shock play important roles.

As investor behaviour may change during a crisis, the dynamics between liquidity variables and jump occurrences may also change. We examine changes by dividing our sample period into pre-crisis, crisis and post crisis periods. However, we do not observe substantial changes in the results in those subsamples indicating the effects of liquidity variables are general on jump occurrences.

Finally, several liquidity variables are shown to contribute to price discovery. Volume, depth, quoted and effective spreads, effective spread shock and order imbalances are significant contributors to the price discovery process. However, the post jump price discovery process does not appear to be very much related with this liquidity variables.

The remainder of the paper is organized as follows. Section 2 documents the liquidity variables used in this paper. A description of data used in this paper and the jump detection methodology are discussed in section 3 and 4 respectively. We present the results in section 5 followed by conclusion in section 6.

2 Variables

Variables used in the liquidity - jump papers

We use the liquidity measures widely used in literature to examine the relationship between market liquidity and stock returns or jumps. We list these liquidity measures and the respective computing methods below:

We measure the immediacy and depth of the market by volume of stocks traded (VOL), depth at best ask price (DPTA) and depth at best bid price (DPTB) as follows:

- (1) $VOL = \text{Number of stocks traded}$
- (2) $DPTA = \text{Best Ask Price} \times \text{Number of Ask}$

$$(3) DPTB = \text{Best Bid Price} \times \text{Number of Bid}$$

Spreads are taken as relative terms and are defined both as quoted spreads (RSPD) and effective spreads (RESPD) as follows:

$$(4) RSPD = (\text{Best Ask Price} - \text{Best Bid Price}) / \text{Mid Quote},$$

$$\text{where Mid Quote} = 1/2 (\text{Best Ask Price} + \text{Best Bid Price})$$

$$(5) RESPD = (\text{Last Price} - \text{Mid Quote}) / \text{Mid Quote}$$

The spread shock variables are shown as the difference between overall relative quoted (effective) spread in an interval and the mean of overall (effective) spread over intervals $t - 5$ to $t - 1$, scaled by its standard deviation. Thus

$$(6) DRSPD = \text{Change in RSPD in time interval } t = \frac{RSPD_t - 1/5 \sum_{j=1}^5 RSPD_{t-j}}{\sigma_{RSPD}} \text{ and}$$

$$(7) DRESPD: \text{Change in RESPD in time interval } t = \frac{RESPD_t - 1/5 \sum_{j=1}^5 RESPD_{t-j}}{\sigma_{RESPD}}$$

The order imbalance variable is defined in our paper as

$$(8) OB (\text{Order Imbalance}) = DPTA_t - DTPB_t$$

Macroeconomic news is considered as the most important causal factor in inciting jumps. Boudt and Petitjean (2014) report that a third of the jumps are caused by macroeconomic news while another 5% of the jumps are associated with firm-specific news. Macroeconomic news tends to effect whole market hence we should observe a market-wide jump as a consequence of arrival of macroeconomic news. We take the jump occurrences in the market index (CNX500) as a proxy of such market-wide jumps.

$$(9) \text{Mkt_Jump} = \text{Jump in the market} - \text{A proxy of arrival in news in the market.}$$

Our dependent variable in the regression analysis is a binary variable indicating the existence of a jump in a given time interval. This variable takes the value of one when the Lee-Mykland jump test detects a jump in a given time interval, otherwise it is zero.

3 Data

We collect data for the Indian market index, stock prices of listed banks from the Thomson Reuters Tisk History (TRTH) database provided by SIRCA. The list of banks is provided in Table 1. We take 10 of the largest banks in term of market capitalization at the end of 2013 in our sample. These 10 banks and financial institutes are listed on the National Stock Exchange (NSE), one of the two major stock exchanges of India. We choose the NSE because it is the largest in the country in terms of daily turnover and number of trades.

Our data covers from the period of January 1, 2004 to December 31, 2013, which includes the global financial crisis in 2008-09. There are a total of 2497 trading days in our sample period. We collect intra-day 15-minute data from TRTH. Dungey, Sayeed and Yao (2016) show through a signature plot that data with 15 minutes sampling frequency for Indian stock prices provides a balance between measurement accuracy and market microstructure noise of high frequency data. We use the last price recorded in each of the 15-minute intervals during the normal trading session of NSE from 9:15 a.m. to 3:30 pm, where missing data are filled with the price of the previous interval which assumes that the price remains unchanged during a non-trading interval. We drop the first 15 minutes of each day to avoid noise associated with market opening. Hence, our trading hours are 9.30 am to 3.30 pm local Indian time. We have 24 observations of 15-minute data for each trading day. We use the CNX500 index, which represents 96.76% of the free float market capitalization of stocks listed on the NSE, as the benchmark market portfolio.

4 Jump Detection Method

Denoting the log price as p , the asset returns process can be depicted as:

$$dp_t = \mu_t dt + \sigma_t dW_t + \kappa_t dq_t, \quad 0 \leq t \leq T, \quad (1)$$

where μ_t represents the drift, σ_t is the diffusion parameter, and W_t is a standard Brownian motion. The third term, $\kappa_t dq_t$ captures the jumps in the price process, where q_t is a counting process with $dq_t = 1$ if there is a jump occurred at time t , and 0 otherwise. κ_t is the size of the jump at time t . The quadratic variation for the process in (1) is defined as

$$QV^{[0,T]} = \int_0^T \sigma_s^2 ds + \sum_{0 < s \leq T} \kappa_s^2. \quad (2)$$

Andersen et al. (2007) (ABD) and Lee and Mykland (2008) (LM) develop tests that can detect jumps at individual return observation level instead of a given time span. They propose a test statistic by calculating the ratio of the return at each observation and a local volatility measure. ABD and LM use different distributions under the null hypothesis of no jumps. The statistics L_i that tests for a jump at every intraday period t_i is defined as

$$L_i = \frac{\Delta_{t_i} p}{\hat{\sigma}(t_i)}, \quad (3)$$

where $\Delta_{t_i} p$ is the intraday return and $\sigma(t_i)$ is the local volatility measure. Here we use the bipower variation of K observations preceding the relevant observation. Thus the test can identify the presence of jumps in an observation against the volatility in the prior period and can be defined as

$$\hat{\sigma}(t_i)^2 = \frac{1}{K-2} \sum_{j=i-K}^{i-1} |\Delta_{t_j} p| |\Delta_{t_{j-1}} p|. \quad (4)$$

Lee and Mykland (2008) suggest a window size K between $252 \times n$ and $\sqrt{252 \times n}$, where n is the number of observations in a day so that the window size poses a balance between being a jump-robust volatility measure and being effective in scaling the trend in volatility. In our paper, we use 80 as values of K for our 15-minute sampling interval where $n=24$.

The asymptotic distribution of the test statistic is as follows:

$$\frac{\max(L_i) - C_n}{S_n} \rightarrow \xi, \quad (5)$$

where $P(\xi \leq x) = \exp(-e^{-x})$, the constants

$$C_n = \frac{(2\log n)^{1/2}}{c} - \frac{\log \pi + \log(\log n)}{2c(2\log n)^{1/2}}, \quad (6)$$

$$S_n = \frac{1}{c(2\log n)^{1/2}}, \quad (7)$$

and

$$c = \sqrt{\frac{2}{\pi}} \quad (8)$$

Thus the LM test detects jumps by comparing the maximized value of L_i to the critical value from the Gumbel distribution. ABD, on the other hand, propose comparing L_i to a normal threshold as L_i is asymptotically normal. By applying their method, jump is identified when $L_i > \Phi_{1-\beta/2}$ where $\beta = 1 - (1 - \alpha)^{\delta^n}$ for a given nominal daily α . We use the LM test method in this paper by applying a significance level of 0.01.

5 Empirical Analysis

We study the dynamics of different liquidity variables such as trading volume, number of trades, depth of ask and depth of bid, the quoted spread ratio and effective spread ratio, the order imbalance in relation to jump occurrences. Further, for each time interval of the aggregate event window we use a Mann-Whitney test to evaluate whether the median value of the liquidity measures on jump intervals is the same on intervals without jumps. We subsequently carry out a parametric analysis to assess the role of liquidity shocks in intra-day jump occurrences and price discovery. Lastly we analyse the impact of economic crisis on the liquidity-jump relationship by comparing sub-sample regression results.

5.1 Liquidity around jumps

To see how liquidity variables behave around the jump time, we take the average values of the liquidity variables within the jump intervals, 15, 30, 45 and 60 minutes prior to the jump times, and also 15, 30, 45 and 60 minutes after the jump times. We remove all the jumps that we find during the first and last trading hours so that we have the liquidity measures 1 hour before and after all detected jumps. In addition we also discard the jumps when we find two consecutive jumps within a two hour interval to avoid liquidity measures which may have overlapping effects from more than one jump.

5.1.1 Return around jumps

Figure 1 shows the average return behaviour around the time intervals when jumps occur. Jump returns are prominently higher than the returns before and after the jump intervals as expected. Panel (a) shows that returns are moving towards zero from positive figures before the jump occurrences. After the jump interval returns become negative and then again start moving toward zero from this point. But if we separate positive and negative jumps in panel (b) and (c) we observe a specific pattern in changes of returns before and after jump intervals. Returns are gradually decreasing before positive jumps, and there is a price correction immediately after the jump return. Returns behave in just the opposite way around the negative jumps. The movement of returns support the overreaction of investors and return reversals hypotheses that have been well documented in the literature (Bondt and Thaler 1985 and Chopra et al. 1992).

5.1.2 Trading volume and depth around jumps

The demand for immediate execution increases around jumps. We observe this evidence in terms of trading volume and number of trades. Figure 2 shows a substantial surge in trading volume, averaged over the sample period at the time of a price jump. In the 15-minute jump interval, the abnormal average volume is three times the median value across the non-jump time intervals. That is consistent with hypothesis that jumps are caused by market inability to absorb new orders

without moving the price significantly up or down. These findings support that jumps are driven by variations in the demand for immediacy. The news alone can not explain the jump arrivals, the impact may be transmitted through the market liquidity changes which in turn move the price up and down.

Our figure also shows that volume decreases gradually from one hour before the jump interval, indicating that the informed investors restrain themselves from trading and wait for the arrival of news. They do get involved in trading once the news has arrived thus contributing a surge in the trading volume. During the one hour following the jump, the volume decreases gradually but remains at a level higher than one hour prior to the jump. This may result from the fact that a section of the investors take time to react after the news become public, or that investors wait to observe the market reaction of the news and then start trading. Our results show a slightly decreasing trend of volume measurements leading to the jump time whereas Boudt and Petitjean (2014) show that the graph is pretty flat until right before the jump. This suggests that the information content of news leading to jumps is essentially unexpected.

The drivers behind the volume surge may be a higher number of trades or larger trade size. Figure 3 and 4 show that both the number of trades and average trade size increase. Nonetheless, the rise of average trade size during the jump intervals is more prominent than the number of trades. When news arrives in the market, informed investors become engaged in trading in higher volume per trade, surging the total volume for the stock and creating jumps. This finding is in line with Kim and Wu (2008) and Easley and O'Hara (1987) who point out that informed traders prefer to trade in larger sizes. Our findings differ slightly from Boudt and Petitjean (2014) who show that both number of trade and average trade size contributed equally to the rapid surge of volume at the jump time. The subtle difference may arise because of the difference between the U.S. market from the Indian market where information may not spread to a wide number of investors as quickly as in the U.S. market.

Figures 6 and 5 illustrates how the market depth at the best bid and ask price increases around jumps. Despite higher spreads and trading volume, quoted market depth is not withdrawn from

the market. Liquidity provision even improves when a jump occurs. It is noteworthy that a greater demand for liquidity rather than a weak liquidity supply, is associated with extreme price changes as noted by Boudt and Petitjean (2014). An increased demand for immediacy does not result in a weak liquidity supply.

5.1.3 Spread around jumps

Figures 7 and 8 shows a significant decrease in average spreads at the time of price jumps indicating a fall of one of the three typical cost components of the spread: order processing costs, inventory holding costs, and/or asymmetric information risk. The effective spread, RESPD is increasing in the hour before jumps and then shows a slump just at the jump interval and stays low for around 45 minutes. The reduction in spread may result from the increase in the volume in the market around jumps. The result is opposite to that for the U.S. market as shown by Boudt and Petitjean (2014), which they explain as a consequence of having the unusually large trading volume at jump time putting pressure on the specialist's or market maker's inventory. As the Indian market is broker driven rather than dealer driven, the impact of the rise in volume may have different impact on spread. The higher volume increases market efficiency and thus decreases the spread cost.

The spread shows an increasing trend one hour before the positive jump arrival time. We may relate this trend with the decrease in volume at the corresponding time intervals. The gradual fall in volume and rise in spread may indicate that investors adjust their behaviour in advance of the release of important news, unlike the U.S. investors.

The impact of jumps on ex-ante liquidity as measured by RSPD is less pronounced as we do not observe any clear fall or rise in RSPD in the jump interval from the previous few intervals. Ex-post liquidity is thus more severely affected than ex-ante liquidity confirming that liquidity providers respond to the occurrences of jumps by being reluctant to improving their displayed quoted prices rather than by posting wider bid-ask spreads. However, the post jump quoted spread rises dramatically when the jump is negative and remains there for the whole hour that we examine indicating that after a large negative return the buyers and sellers become overcautious

in quoting their respective prices. We do not observe any such pattern in quoted spread after a positive jump occurrence.

5.1.4 Order imbalance

Figure 9 shows the trading activity at the time of a jump occurrence is unbalanced. Positive (negative) jumps are characterized by large buying (selling) pressure as we observe sharp step up in the depth imbalances in the jump intervals. According to Chordia et al. (2002) the degree of order imbalance is an indication about the relevance of the information released to the market. Since a high number of trades may be considered as a large sample of independent observers of the same signal, a high disequilibrium between buy and sell transactions suggests that the market participants widely agree on the positive or negative effect of news.

Our results show that the imbalance measure drops back quickly to its pre-jump level after the jump in the case of negative jumps. In contrast, trading imbalances are more persistent after the positive jumps. Although market participants may initially agree on the direction of the prices, heterogeneous beliefs about the revised fundamental value of the asset appear among the investors afterwards; this is more pronounced for negative jumps, leading to a more balanced interaction between bulls and bears. It also indicates that the overreaction to the news is more prevalent for negative jump returns than the positive ones.

5.1.5 Mann-Whitney Test Results

The Mann-Whitney test is used to rank a series from smallest value to largest, and to compare the sum of the ranks from one subgroup to the sum of the ranks from another subgroup. If the groups have the same median, the values should be similar. We use EViews to reports the asymptotic normal approximation to the U-statistic and the p-values for a two-sided test; for details, see Sheskin (2003). The test in Eviews is based on a one-way analysis of variance using only ranks of the data.

Table 3 reports the Mann-Whitney test results of whether the liquidity measures in the time

interval of jump occurrences come from the same distribution as the ones in intervals without jumps for the ten sample banks. We show the results for all jumps, positive jumps and negative jumps in separate rows for the liquidity variables used in this paper. Volume is significantly different during the jump intervals for all ten banks at the 95% confidence level and for both positive and negative jumps, which shows that jump occurrences are strongly related with trading volumes. For example, the median volume of non-jump intervals is 28,335 while the median volume of jump intervals is 30,699 for HDBK bank. The arrival of news prompts investors to trade in the market, and the price changes rapidly depending on the strength of buying or selling pressure. Median values of both the depth measures DPTB and DPTA (depth is bid and ask) are also significantly higher in the jump intervals, supporting the notion of the sudden change in stock prices backed by investors' desire for immediacy.

The relative spread measures the quoted spread, RSPD and the effective spread, RESPD, are found to have significantly different values in the jump and non-jump intervals. For HDBK Bank the median value of RSPD .000854 during the jump intervals which is around twice as large as the median value of .000425 in non-jump intervals. We see similar patterns for the relative effective spread measure, RESPD showing that extreme price movements are associated with increased transaction costs for the investors.

The order imbalance measure, OB also varies significantly in jump and non-jump intervals. The median order imbalance is almost four times higher in jump intervals than the non-jump intervals for HDBK Bank. The Mann-Whitney test results show that the difference is more prominent for negative jump intervals than for positive jump intervals. We can interpret this as evidence that the extreme price drops are connected with high selling pressure which creates substantial order imbalance in the market.

Our results are consistent to the findings of Boudt and Petitjean (2014) and Jiang et al. (2011). Though the Indian market is order driven rather than quote driven, our results reveal that the liquidity variables are significantly related to the jump occurrences as in the USA market.

5.2 Impact of Liquidity on jump probabilities

The empirical findings documented in the previous subsections suggest that liquidity shocks around jumps are substantial. In this subsection, we examine how shocks in volume, depth, effective and quoted spreads and order imbalance variables explain the probability that a jump occurs. The following regression is estimated for each of the 10 banking stocks over the sample period from January 1 2004 to December 31, 2013:

$$P(Jump_j = 1 | X) = G(\alpha_j + \beta_1 MJ_j + \beta_2 Vol_j + \beta_3 DPTB_j + \beta_4 RSPD_j + \beta_5 RESPD_j + \beta_6 OB_j + \beta_7 DDRSPD_j + \beta_8 DDRESPD_j), \quad (9)$$

where X is a vector of regressors. We include a dummy variable, MJ to represent the existence of jumps in the market in a given time interval. We use this variable as a proxy of macroeconomic news arrival with the assumption that the whole market will be affected by the news. We use only one measure of depth, $DPTB$ as the co-relation between $DPTB$ and $DPTA$ is very high. We run two binary dependent variable models, as shown in Table 4 and 5, to test whether any liquidity variable contributes to the occurrence of stock jumps contemporaneously. The probit and logit models lead to the same conclusions. First, we examine the contemporaneous effect of liquidity on overall jumps, positive jumps and negative jumps. We report the average coefficients of the variables in the second column and from 3rd to 6th columns we show the number of stocks where we find the relationship is positive and significant at 5%, positive and significant at 10%, negative and significant at 5% and negative and significant at 10% level for each independent variables. We also report the average McFadden R-squared values for each set of regressions.

Occurrences of market jumps increase the probability of having jumps in the individual stocks significantly at the 5% level for all of the ten stocks analysed in our paper. A jump in the market indicates that the broad investors receive a signal that affects a large number of stocks. The individual stocks are then more likely to react to the signal. The probability of either a positive

or negative jump at the time of occurrence of market jumps both increases.

An increase in volume (VOL) increases the likelihood of contemporaneous jumps for Indian stocks as shown in all ten of our sample stocks. When volume surges the price changes rapidly in the direction that is dictated by the buying vs. selling pressures. We see the impact of volume is more pronounced for positive jumps than negative jumps, and the signs of the coefficients are also opposite. When volume increases, the market takes it as a bullish signal and the corresponding response generates a positive jump. Conversely, in case of negative jumps we find that the reaction is significantly negative in four regressions and positive in only one regression. We may interpret the results in two ways. First, the abnormal negative returns may arise from bad news rather than from the impact of volume surge. Second, it is the decrease in market volume that increases the chances of having negative jumps in the stock. A fall in volume depresses the market, the investors become bearish, and the stock price falls substantially creating negative jumps.

The market depth at bid price (DPTB) has a significant positive coefficient in 4 stocks for positive jumps, and 6 stocks for negative jumps. This shows that the depth is more associated with negative jumps than the positive jumps. The increase in buying pressure implies greater demand for liquidity rather than a weak liquidity supply, resulting in extreme price changes in either direction. However, the depth variable is not as strong as the volume variable in determining the probability of jump occurrences for the Indian banking stocks.

Among the two spread ratios we apply in our paper the quoted spread, RSPD and effective spread, RESPD, we see that RESPD affects the probability of jump occurrences in more stocks (10 vs 8). However we see quite different results in the case of positive jumps and negative jumps. In all 10 regressions the quoted spread significantly affects the probability of jump occurrences at the 5% level, whereas the effective spread is negatively significant at 5% in 6 regressions, and positively significant at 5% level in one regression. In contrast, the effective spread can explain the negative jumps significantly in all ten regressions. The results show that an increase in effective spread can increase the probability of having a negative jump. High effective spreads may be indicative of greater market imbalance between buying and selling pressures. The abnormal negative returns or

jumps are closely related to such market imbalances, which may result from spread of negative news in the market. The negative coefficient of RESPD in 6 stocks in case of positive jumps means that, increase in RESPDs reduces the chance of having an abnormally high positive returns, indicating that positive high returns are not associated with market imbalances rather the higher differences between ask and bid prices.

Changes in the spread ratio in the contemporaneous time intervals from their last five values also significantly affect the probabilities of jump occurrences. Among the two variables, DDRSPD (change in quoted spread) and DDRESPD (change in effective spread), we find DDRESPD has higher impact on jump probabilities. The rapid change in effective spreads has higher likelihood of being accompanied by jump returns either positive or negative.

The liquidity order imbalance variable, OB does not seem to be crucial variable for jumps given that it has a significant coefficient at 5% only in two regressions when we look at the overall jumps. However, by analysing the positive and negative jump regression outcomes we see quite a different story. OB has a significantly negative coefficient in 9 regressions at the 5% level in the case of positive jumps. This variable is positively significant at the 5% level in 9 out of 10 regressions in the case of negative jumps. Thus an increase in order imbalance is associated with an increase in the probability of negative jumps, and a decrease in the probability of positive jumps.

Overall the liquidity measures along with the market jumps can explain 19.17% of the jump probabilities for all jumps as indicated by the average McFadden R^2 . The difference in McFadden R^2 values between the regressions for positive jumps and the regression for negative jumps is noteworthy. While this value is only 11.53% for positive jumps, it is 35% for the negative jumps. Thus liquidity variables are much more strongly associated with negative returns rather than positive returns contrasting with our initial predictions. Our results indicate that abnormal negative price movements are more sensitive to the liquidity shocks. The probit regression results are largely supported by the logit regression results presented in Table 5.

We also examine the lagged effect of the liquidity values on jump occurrences for the Indian banking stocks. Table 6 shows that it is primarily the volume depth variables that have significant

impact on the probabilities of positive jumps. VOL and DPTA have significant positive coefficients in 9 and 7 regressions respectively. In the case of negative jumps it is the effective spread and its variations from previous values (RESPD and DRESPD) that have significant coefficients. The results indicate that increase in market depth leads to a positive jump in the next 15-minute interval, while the spread change points towards negative jumps.

5.3 Liquidity and Global Financial Crisis

We examine whether the liquidity-jump relationship of Indian banks shifts during crisis periods. Traders may react differently to the liquidity shocks during the crisis period than the non crisis periods. The results are reported in Table 7. As the global financial crisis (GFC) of 2007-2009 falls within our sample period, we analyse the impact by taking sub-samples referring to the pre-crisis, crisis and post-crisis periods. We follow the the Dungey et al. (2015) in identifying the crisis period, who identify 3rd July, 2007 as the beginning of the crisis period when the connection between the troubled US sub-prime mortgage market and the global credit markets came to light and just prior to the resignation of the CEO of UBS on 5th July. The second phase of crisis starts from early October 2008, following the bankruptcy of Lehman Brothers and subsequent rescue of the American Insurance Group. The breakpoint from crisis period to post crisis has been identified as 15 May 2009, which is the day of the publication of stress test results and new capital raising by US banks, and the LIBOR rate falling below 1%. In our paper we consider both phases of crisis period as one crisis period. Thus in our sample, 1 January 2004 to 2 July 2007 is considered as the pre-crisis period, 3 July 2007 to 15 May 2009 as the crisis period and 16 May 2009 to 31 December 2013 as the post crisis period.

We do not find any notable change in the liquidity-jump relationship among the pre-crisis, crisis and post-crisis periods. The volume, effective spread ratio and the change in effective spread ratio remain dominant in determining the probability of jumps in all three phases. The average McFadden R^2 has risen from 19.24% to 22.87% from the pre-crisis period to the crisis period, indicating a rise in explanatory power of liquidity variables in the crisis period. However, the

McFadden R^2 of the post-crisis period is 22.47% which is similar to the crisis period.

5.4 Price Discovery and Liquidity

The impact of liquidity variables on stock returns when there are jumps is examined by applying the following mode

$$\begin{aligned}
\Delta_{ti}^n p = & \alpha_j + \beta_1 Jump_{j-1} + \beta_2 Vol_j + \beta_3 Vol_j * Jump_{j-1} + \beta_4 DPTB_j + \beta_5 DPTB_j * Jump_{j-1} \\
& + \beta_6 RSPD_j + \beta_7 RSPD_j * Jump_{j-1} + \beta_8 RESPD_j + \beta_9 RESPD_j * Jump_{j-1} \\
& + \beta_{10} OB_j + \beta_{11} OB_j * Jump_{j-1} + \beta_{12} DDRSPD_j + \beta_{13} DDRSPD_j * Jump_{j-1} \\
& + \beta_{14} DDRESPD_j + \beta_{15} DDRESPD_j * Jump_{j-1} + v_j
\end{aligned} \tag{10}$$

A dummy variable representing the jump with one lag is added to the the model, as well as the interaction of this dummy variable with all the liquidity variables. We can see from Table 8 that the jump dummy variable with a lag is positively significant at 5% in only 3 out of 10 regressions. Hence, we do not find strong evidence that a jump in a given time period has forecasting value for the return prediction of the next period. Similarly none of the interaction variables with the jump dummy are overwhelmingly significant in these 10 regressions. Volume itself significantly impacts the contemporaneous return (10 out of 10 regressions at 5% level) but the interaction with the dummy variable with one lag is negatively significant in 5 out of 10 regressions. When we have a jump in one period the volume has a reverse relationship with the return. When jumps increase the uncertainty in the market, investors want to sell their assets immediately, the price drops and buyers are attracted to the respective stock to take advantage of low prices, thus volume rises in the market. The quoted spread, RSPD, has a positive and significant (at 5% level) impact on the return, indicating that an increased difference in bid and ask is associated with higher returns. The effective spread, RESPD, on the other hand is negatively associated with the return. When the executed price moves away from the mid quote, the return decreases. However, the interaction terms of these two spread variables with the jump dummies are not significant in most of the 10

regressions. The order imbalance variable, OB, is negatively significant in the regressions while the interaction term with jump is not significant in any case.

Among the two change in spread variables, the change in effective spread, DDRESPD, is found to be negatively significant in determining the return in 8 out of 10 regressions whereas the other variable, change in quoted spread, DDRSPD, in only two regressions. Hence it is the effective spread rather than the quoted spread that changes with the return. However, when we incorporate the jump of the previous interval with these variables, none of the variables shows any definite relationship with stock returns.

Overall, it seems that having jumps in the previous time interval does not have a notable impact on the liquidity measures in explaining the variations in stock returns. However, the liquidity measures themselves are found to have significant coefficients in the return regressions. The average adjusted R^2 value of these regressions are 36.90%.

6 Conclusion

Our paper examines the relationship of liquidity variables with the jump movements of equity securities. Using ten of the largest Indian banking stocks we explore the relationship of liquidity variables with the jump returns of these stocks.

We use the jump test developed by Lee and Mykland (2008) in his study and observe jumps in 1.58% to 1.73% return observation in ten banking stocks. We implemented an event study method to analyse the link between intra-day liquidity dynamics and properly identified intra-day jumps regardless of the arrival of information. To complement the non-parametric event study we apply a parametric analysis to assess the contribution of liquidity shocks in both the occurrence of jumps and the price discovery process. We find notable variations in the liquidity measures around the jump intervals in the stocks. If we characterize liquidity by market spread, trading quantity and immediacy, market liquidity conditions do improve around jumps. We examine the drivers behind this volume surge and find that the rise in average trade size is more prominent than the rise in

number of trade. We show that the negative jumps are associated with higher bid and ask depth. The results indicate that the demand of immediacy of traders may cause jumps in stock returns. The Mann-Whitney test results also confirm the significant changes in the liquidity variable during the jump intervals from non-jump intervals.

We examine the contemporaneous and lagged effects of liquidity factors on the probability of jump occurrences. We find evidence of both effects in different degrees for jump occurrences. Our probit estimation shows that the explanatory power of contemporaneous liquidity variables is much higher than the lagged variables for all jumps. When we divide the sample into negative and positive jumps we observe that the liquidity variables have more explanatory power in determining probability of negative than that of positive jumps. It seems that negative jumps in market activities require changes in market activities than the positive jumps to occur. Among the liquidity variables, volume, quoted spreads, effective spread shock and order imbalance are key drivers behind the occurrences positive jumps. For negative jump occurrences the effective spread, effective spread shock play important roles.

We extend the analysis by dividing our sample into pre-crisis, crisis and post-crisis period to examine changes in the dynamics between liquidity variables and jump occurrences during the crisis period. We do not observe substantial changes in the results in those subsamples indicating that the effects of liquidity variables are general.

Finally, several liquidity variables are shown to contribute to price discovery. Volume, depth, quoted and effective spreads, effective spread shock and order imbalances are significant contributors in the price discovery process. However, the post jump price discovery process does not seem to be strongly related to the liquidity variables.

The Tables

Table 1: List of Sample Banks

We have 41 Indian banks listed on the National Stock Exchange (NSE) at the end of 2013. Ten largest banks in market capitalization shown below have been analysed in this paper. The first column shows the Codes used in TRTH database to identify the banks. The second column shows the name and the third column reports the market capitalization of each bank in thousand Indian Rupees (INR).

Codes	Banks	Market Cap.
ADBK	Andhra Bank	52,880,344
HDBK	HDFC Bank	1,484,995,359
SBI	State Bank of India	1,390,909,091
ICBK	ICICI Bank	1,205,864,240
AXBK	Axis Bank	608,925,751
KTKM	Kotak Mahindra Bank	485,807,555
BOB	Bank of Baroda	278,659,424
PNBK	Punjab National Bank	243,360,705
INBK	Indusind Bank Limited	211,649,245
BOI	Bank of India	180,260,454
CNBK	Canara Bank	170,311,355

Table 3: Mann-Whitney Results for Liquidity measures

Here the Mann-Whitney test results of Liquidity variable between jump and nonjump time intervals are shown for all ten banks. The reported values are Wilcoxon/Mann-Whitney test statistics. The asterisks *, **, and *** indicate that the null hypothesis of equality of the median values of the variables across different subsamples are rejected at the significance level of 10%, 5%, and 1% level, respectively.

Variables	Jumps	HDBK	SBI	ICBK	AXBK	KTKM	BOB	PNBK	INBK	BOI	CNBK
Volume	Overall	4.27***	32.76***	16.82***	11.97***	11.68***	28.03***	27.75***	28.96***	32.68***	25.55***
	Positive	5.20***	24.41***	15.34***	10.22***	10.87***	23.84***	23.85***	27.59***	28.18***	22.75***
	Negative	0.78***	20.30***	9.05***	5.57***	4.84***	15.09***	14.29***	11.84***	17.05***	12.43***
RSPD	Overall	16.84***	12.68***	4.21***	17.28***	14.65***	13.09***	6.49***	14.67***	6.16***	7.87***
	Positive	6.15***	6.08***	6.90***	8.87***	5.85***	4.27***	3.65***	5.58***	4.07***	4.27***
	Negative	9.31***	3.79***	8.67***	6.82***	6.41***	5.04***	5.60***	6.13***	4.64***	7.01***
RESPD	Overall	3.38***	3.99***	4.21***	3.38***	1.4	2.64***	21.07***	4.14***	20.59***	20.13***
	Positive	4.09***	5.37***	5.81***	7.55***	4.38***	4.72***	6.02***	10.12***	6.47***	5.71***
	Negative	9.31***	22.81***	24.54***	22.94***	19.91***	23.96***	24.65***	22.25***	23.61***	23.90***
DPTA	Overall	1.74*	3.59***	3.27***	7.92***	6.95***	4.13***	3.25***	5.70***	4.70***	5.16***
	Positive	10.13***	7.90***	8.82***	11.08***	8.27***	10.87***	6.47***	8.19***	7.16***	9.89***
	Negative	8.84***	5.66***	7.48***	5.57***	5.29***	6.94***	5.13***	7.41***	3.85***	8.09***
DPTB	Overall	13.76***	30.88***	12.2	12.94***	10.02***	13.06***	9.09***	11.73***	8.52***	13.10***
	Positive	9.92***	7.39***	8.65***	10.93***	8.20***	10.59***	6.15***	8.44***	7.09***	9.58***
	Negative	9.30***	6.44***	8.49***	6.47***	5.83***	7.64***	6.65***	7.94***	4.76***	8.82***
OB	Overall	2.30**	1.03	4.7***	3.53***	2.29**	2.06**	4.23***	4.50***	4.77***	1.43
	Positive	2.46**	5.74***	1.57	1.97***	1.43	3.26***	5.48***	1.13	0.96	3.80***
	Negative	5.52***	7.76***	8.48***	7.66***	5.03***	6.66***	12.40***	5.55***	8.27***	6.52***

Table 2: Summary Statistics of Jumps in ten banking stocks

The 2nd column shows percentage of jump returns in total observations. In the fourth and 6th column we report the percentage of positive and negative jumps in total number of jumps respectively. Total Number of return observations: 87672. Jumps are measured using Lee and Mykland (2008) method.

Banks	Total Jumps	Percentage of total obs.	Positive jumps	Percentage of total jumps	Negative Jumps	Percentage of total jumps
HDBK	1454	1.66%	761	52.34%	693	47.66%
SBI	1382	1.58%	742	53.69%	640	46.31%
ICBK	1465	1.67%	776	52.97%	689	47.03%
AXBK	1520	1.73%	869	57.17%	651	42.83%
KTKM	1453	1.66%	833	57.33%	620	42.67%
BOB	1421	1.62%	779	54.82%	642	45.18%
PNBK	1496	1.71%	827	55.28%	669	44.72%
INBK	1486	1.69%	857	57.67%	629	42.33%
BOI	1470	1.68%	825	56.12%	645	43.88%
CNBK	1468	1.67%	832	56.68%	636	43.32%
Average	1461.5	1.67%	810.1	55.41%	651.4	44.59%

Table 4: Regression Results: Jumps vs Liquidity - Probit

The probit regression results estimated on ten bank data separately are summarized here. The dependent variable is Jump occurrences, a binary variable, the independent variables are as described in Equation 9. The average coefficients are shown in the second column. Column three and four report the number of regressions where the positive coefficients are significant at 5% and 10% level respectively and while column five and six show the number of negative coefficients significant at 5% and 10% level respectively. The results are reported for all jumps in the first panel, for positive jumps in the second panel and for negative jumps in the third panel.

Variables	Coefficients	Positive at 5%	Positive at 10%	Negative at 5%	Negative at 10%
All Jumps					
C	-2.7217	0	0	10	0
JUMP_MKT	1.4435	10	0	0	0
VOL1	0.0000	10	0	0	0
DPTB	0.0000	4	0	0	0
RSPD	60.5248	8	0	0	0
RESPD	66.2902	10	0	0	0
OB	0.0000	0	0	2	2
DDRESPD	0.0521	9	1	0	0
DDRSPD	-0.0004	1	0	3	0
McFadden R-sq.	0.1917				
Positive Jumps					
C	-2.7216	0	0	10	0
JUMP_MKT	1.0295	10	0	0	0
VOL1	0.0000	10	0	0	0
DPTB	0.0000	4	0	0	0
RSPD	150.7759	10	0	0	0
RESPD	-17.9099	1	0	6	0
OB	-0.0000	0	0	9	1
DDRESPD	0.0375	9	0	0	0
DDRSPD	-0.0007	0	0	5	0
McFadden R-sq.	0.1153				
Negative jumps					
C	-3.2497271	0	0	10	0
JUMP_MKT	1.3633	10	0	0	0
VOL1	-0.0000	1	0	4	0
DPTB	0.0000	6	0	0	0
RSPD	16.4809	3	1	1	0
RESPD	102.0840	10	0	0	0
OB	0.0000	9	0	0	0
DDRESPD	0.0534	10	0	0	0
DDRSPD	-0.0001	0	0	3	0
McFadden R-sq.	0.3500				

Table 5: Regression Results: Jumps vs Liquidity - Logit

The logit regression results estimated on ten bank data separately are summarized here. The dependent variable is Jump occurrences, a binary variable, the independent variables are as described in Equation 9. The average coefficients are shown in the second column. Column three and four report the number of regressions where the positive coefficients are significant at 5% and 10% level respectively and while column five and six show the number of negative coefficients significant at 5% and 10% level respectively. The results are reported for all jumps in the first panel, for positive jumps in the second panel and for negative jumps in the third panel.

Variables	Coefficients	Positive at 5%	Positive at 10%	Negative at 5%	Negative at 10%
All Jumps					
C	-5.3756	0	0	10	0
JUMP_MKT	2.9173	10	0	0	0
VOL1	0.0000	9	1	0	0
DPTB	0.0000	4	2	0	0
RSPD	120.2242	8	0	0	0
RESPD	140.3423	10	0	0	0
OB	-0.0000	0	0	2	1
DDRESPD	0.1080	10	0	0	0
DDRSPD	-0.0008	0	0	4	0
McFadden R-sq.	0.1798				
Positive Jumps					
C	-5.5192	0	0	10	0
JUMP_MKT	2.3664	10	0	0	0
VOL1	0.0000	10	0	0	0
DPTB	0.0000	4	0	0	0
RSPD	333.8768	10	0	0	
RESPD	-40.0506	1	0	4	0
OB	-0.0000	0	0	9	1
DDRESPD	0.0771	9	1	0	0
DDRSPD	-0.0013	0	0	5	0
McFadden R-sq.	0.1032				
Negative jumps					
C	-6.7348	0	0	10	0
JUMP_MKT	2.9808	10	0	0	0
VOL1	-0.0000	1	0	4	0
DPTB	0.0000	6	0	0	0
RSPD	27.5518	4	0	1	0
RESPD	220.2321	10	0	0	0
DDRESPD	0.1142	10	0	0	0
DDRSPD	-0.0001	0	0	0	1
McFadden R-sq.	0.3240				

Table 6: Regression results with 1 lag liquidity variables

The probit regression results estimated on ten bank data separately are summarized here. The dependent variable is *Jump* occurrences, a binary variable, the independent variables are as described in Equation 9 in one order lagged form. The average coefficients are shown in the second column. Column three and four report the number of regressions where the positive coefficients are significant at 5% and 10% level respectively and while column five and six show the number of negative coefficients significant at 5% and 10% level respectively. The results are reported for all jumps in the first panel, for positive jumps in the second panel and for negative jumps in the third panel.

Variables	Coefficients	Positive at 5%	Positive at 10%	Negative at 5%	Negative at 10%
All Jumps					
C	-2.2233			10	
JUMP_MKT	0.8111	10			
VOL1	0.0000	7			
DPTB	-0.0000			7	1
RSPD	17.2791	2	1		
RESPD	10.8049	4	1		
OB	-0.0000			2	
DDRESPD	0.0165	6	2		
DDRSPD	-0.0002			2	2
McFadden R-sq.	0.0154				
Positive Jumps					
C	-2.6401			10	
JUMP_MKT	0.7594	8	1		
VOL1	0.0000	9			
DPTB	-0.0000	7	1		
RSPD	25.5296	3			
RESPD	-3.2890		1	1	
OB	-0.0000			5	
DDRESPD	0.0196		5		
DDRSPD	-0.0006			2	1
McFadden R-sq.	0.0129				
Negative jumps					
C	-2.8059			10	
JUMP_MKT	0.9100	10			
VOL1	0.0000	2			
DPTB	-0.0000			3	
RSPD	10.0960	2			
RESPD	22.1471	9			
OB	0.0000	2			
DDRESPD	0.0155	5			
DDRSPD	-0.0002			5	1
McFadden R-sq.	0.0217603				

Table 7: Regression results to showing the impact of GFC

The probit regression results estimated on ten bank data separately are summarized here. The dependent variable is *Jump* occurrences, a binary variable, the independent variables are as described in Equation 9. The average coefficients are shown in the second column. Column three and four report the number of regressions where the positive coefficients are significant at 5% and 10% level respectively and while column five and six show the number of negative coefficients significant at 5% and 10% level respectively. The results are reported for pre-crisis period in the first panel, for crisis period in the second panel and for post crisis period in the third panel.

	Coefficients	Positive at 5%	Positive at 10%	Negative at 5%	Negative at 10%
Pre-crisis					
C	-2.5015			10	
JUMP_MKT	1.3289	10			
VOL1	0.0000	10			
DPTB	0.0000			5	
RSPD	15.4059	3			
RESPD	76.5011	10			
OB	0.0000	1	1	1	
DDRESPD	0.0575	9			
DDRSPD	0.0049	1		1	
McFadden R-sq.	0.1924				
Crisis					
C	-2.7774			10	
JUMP_MKT	1.4725	10			
VOL1	0.0000	8	1		
DPTB	0.0000			7	2
RSPD	-81.1604			2	
RESPD	76.3678	10			
OB	0.0000	1		1	
DDRESPD	0.0784	10			
DDRSPD	0.0073			3	
McFadden R-sq.	0.2287				
Post-crisis					
C	-3.3282			10	
JUMP_MKT	1.6466	10			
VOL1	0.0000	8	1		
DPTB	0.0000	8			
RSPD	74.4250	7	1		
RESPD	117.2170	10			
OB	0.0000			1	2
DDRESPD	0.0534	10			
DDRSPD	-0.0002			2	
McFadden R-sq.	0.2247				

Table 8: Regression of price discovery

The OLS regression results estimated on ten bank data separately are summarized here. The dependent variable is stock return, the independent variables are as described in Equation 10. The average coefficients are shown in the second column. Column three and four report the number of regressions where the positive coefficients are significant at 5% and 10% level respectively and while column five and six show the number of negative coefficients significant at 5% and 10% level respectively.

Variables	Coefficients	Positive at 5%	Positive at 10%	Negative at 5%	Negative at 10%
C	0.0011	10			
JUMP01(-1)	0.0006	3	1		
VOL1	1.5956E-08	10			
VOL1*JUMP01(-1)	-7.208E-09			4	1
DPTB	4.8904E-10	5		5	
DPTB*JUMP01(-1)	-1.48723E-09				
RSPD	0.9433	10			
RSPD*JUMP01(-1)	0.7797	3			
RESPD	-0.8721			10	
RESPD*JUMP01(-1)	0.1210	1	2		
OB	-7.853E-09			10	
OB*JUMP01(-1)	-2.72636E-09				
DDRESPD	-0.0001			8	
DDRESPD*JUMP01(-1)	2.3E-06	2	1	1	1
DDRESPD	-2.917E-08			2	1
DDRESPD*JUMP01(-1)	-2.21449E-06			4	
Adjusted R-sq.	0.3690				

The Figures

Figure 1: Return around the jump observations

The average returns at 15, 30, 45 and 60 minutes before and after jump times are shown here.

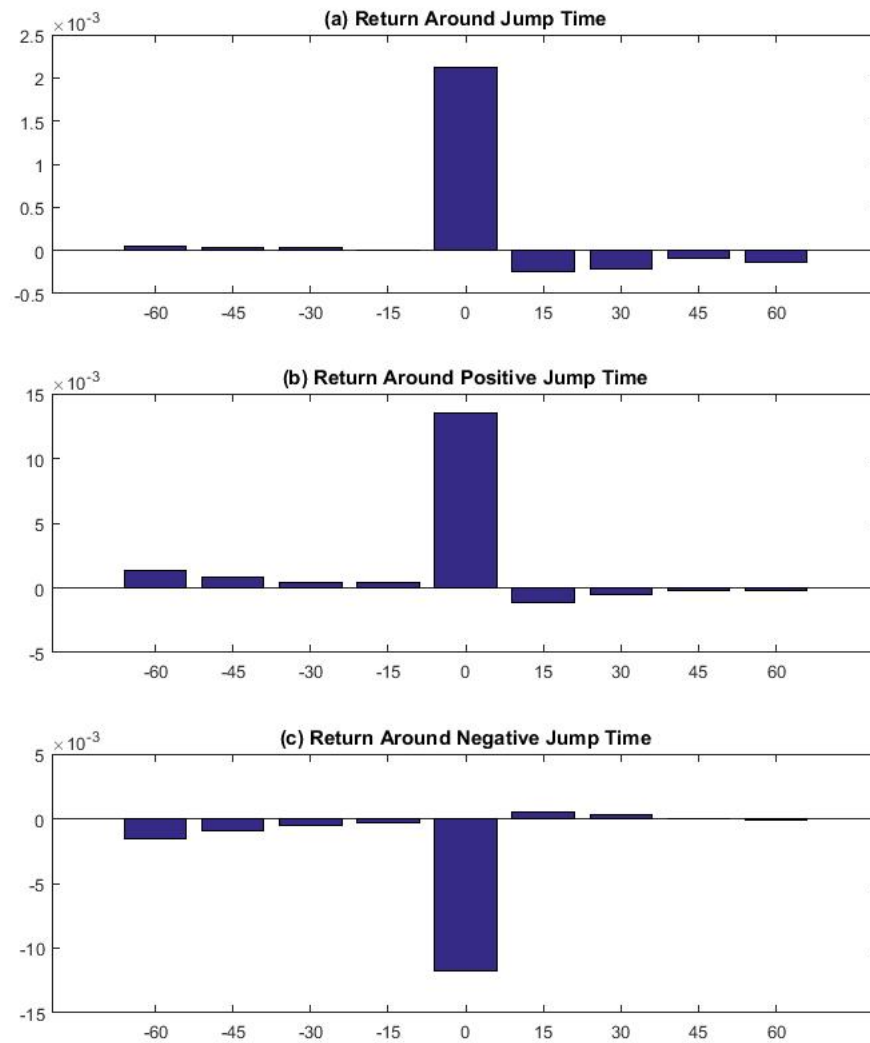


Figure 2: Volume around the jump observations

The average trade volumes at 15, 30, 45 and 60 minutes before and after jump times are shown here.

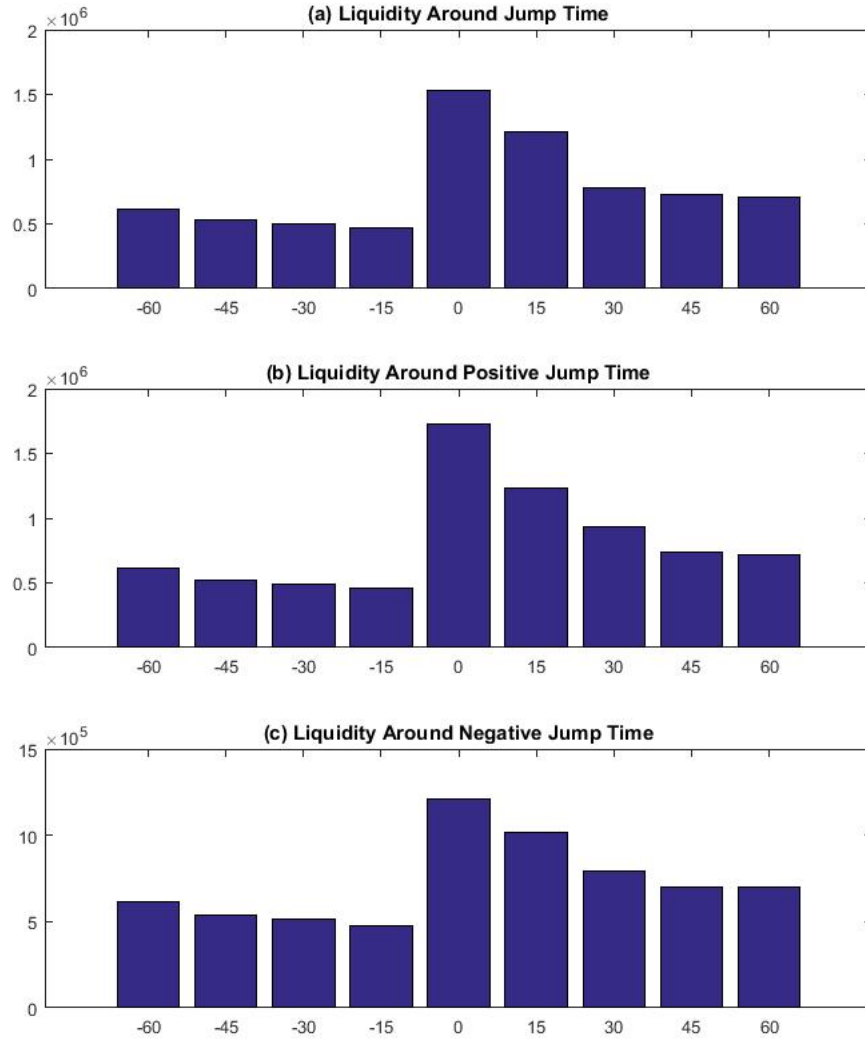


Figure 3: Number of trade around jump observations

The average number of trades at 15, 30, 45 and 60 minutes before and after jump times are shown here.

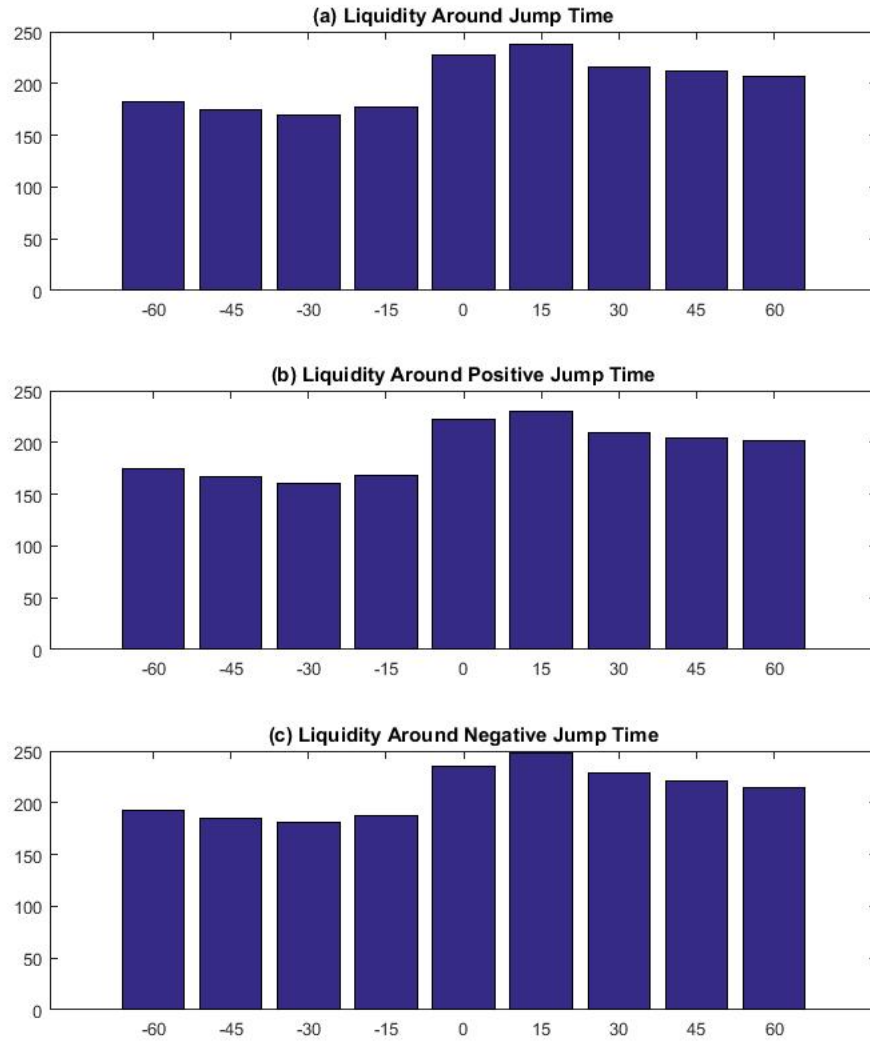


Figure 4: Average Trade Size

The average trade sizes at 15, 30, 45 and 60 minutes before and after jump times are shown here.

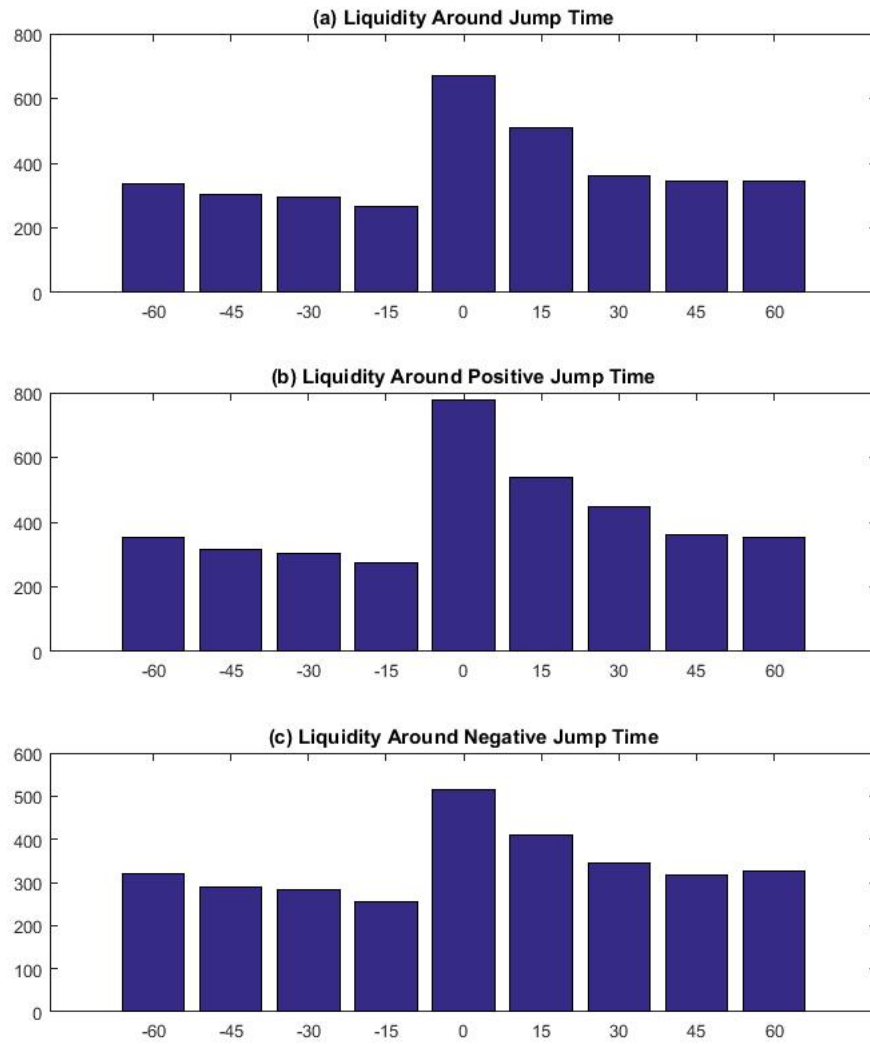


Figure 5: Depth of ask around jump observations

The average depth of ask prices at 15, 30, 45 and 60 minutes before and after jump times are shown here.

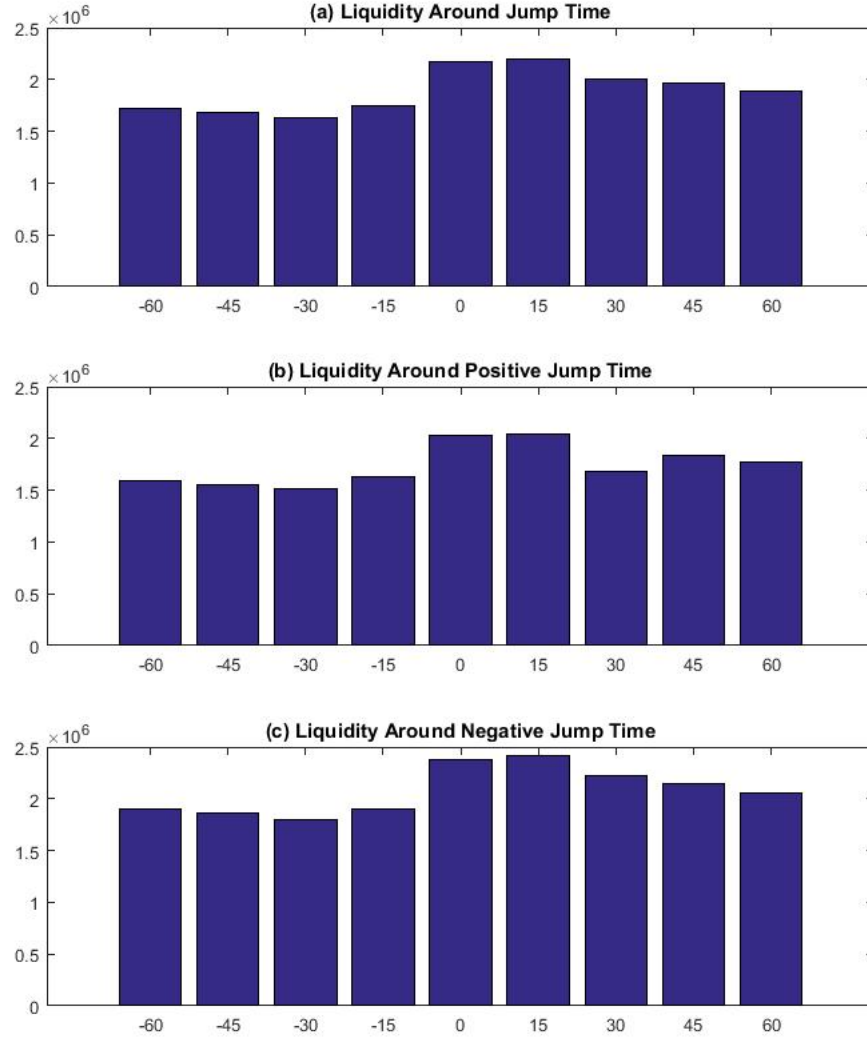


Figure 6: Depth of bid around jump observations

The average depth of bid prices at 15, 30, 45 and 60 minutes before and after jump times are shown here.

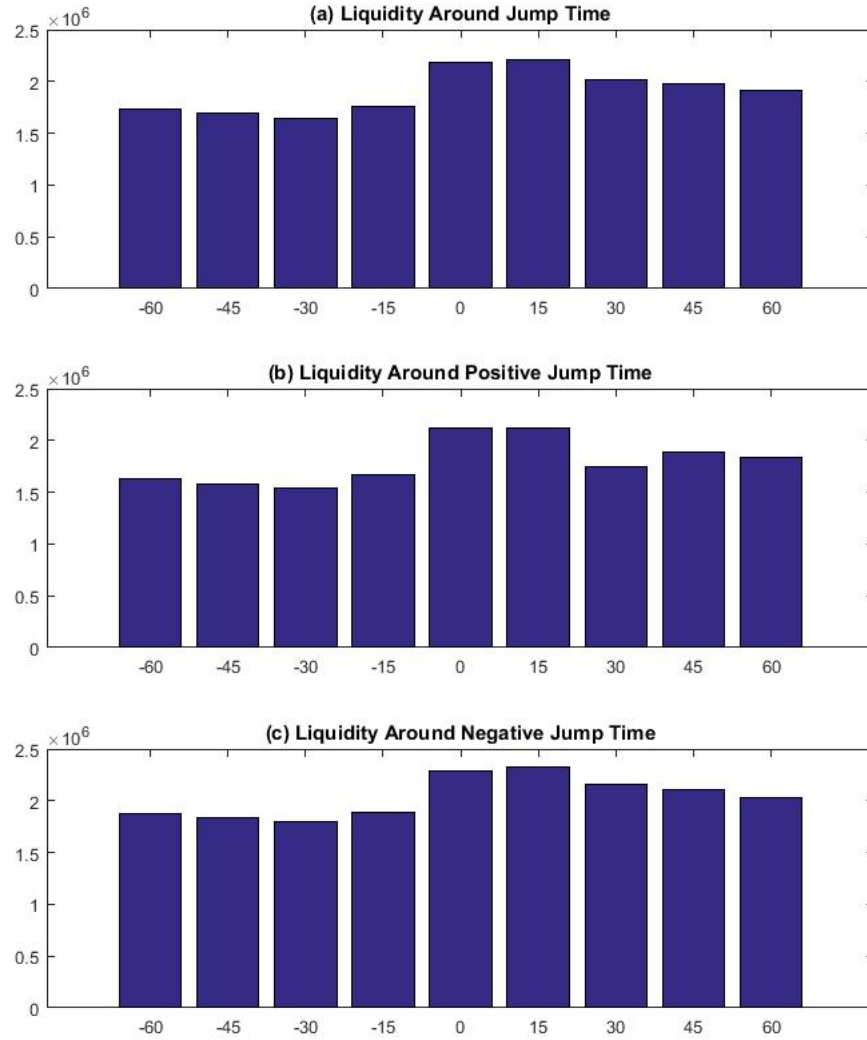


Figure 7: RSPD around the jump observations

The average quoted spread (RSPD) ratios at 15, 30, 45 and 60 minutes before and after jump times are shown here.

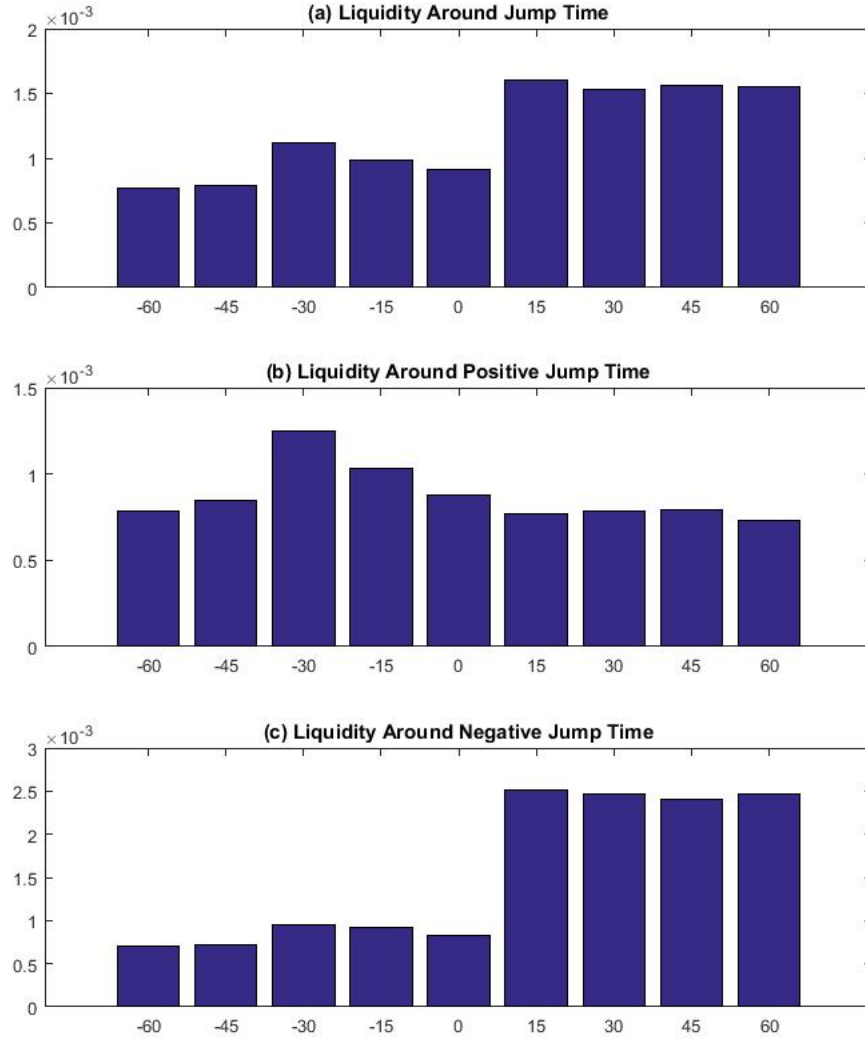


Figure 8: RESPD around the jump observations

The average effective spread ratios (RESPD) at 15, 30, 45 and 60 minutes before and after jump times are shown here.

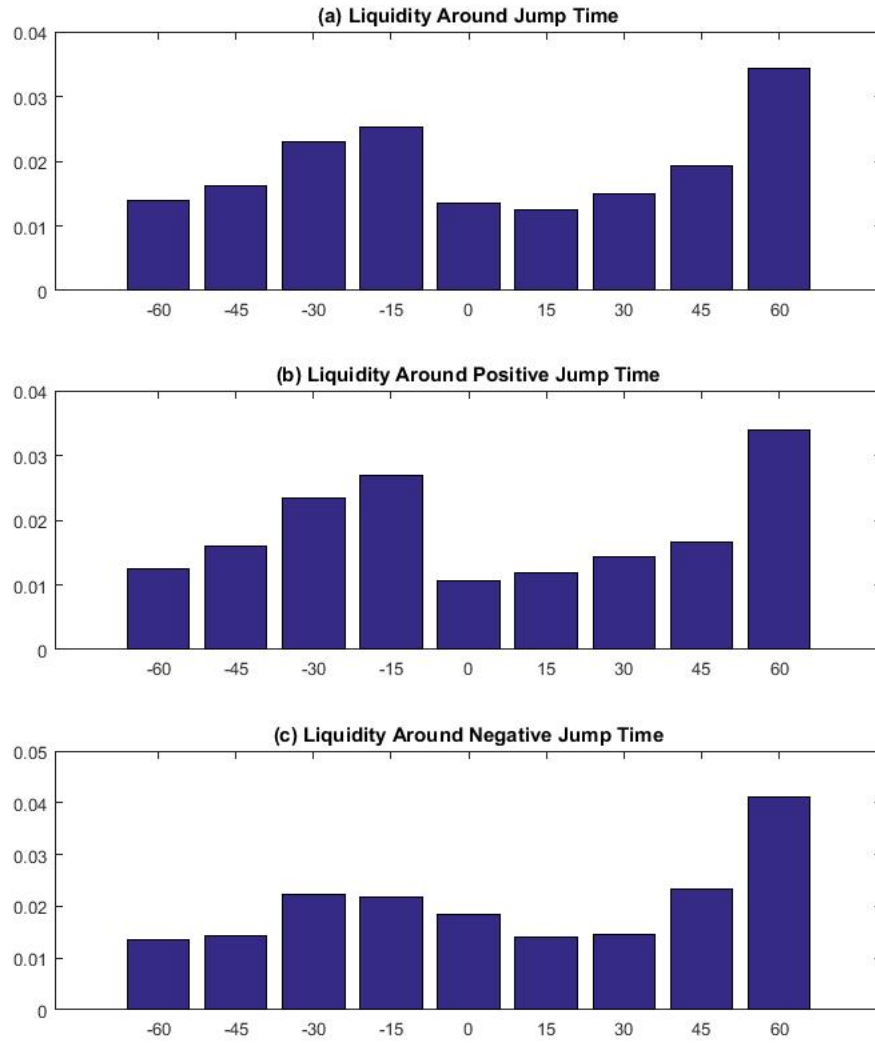
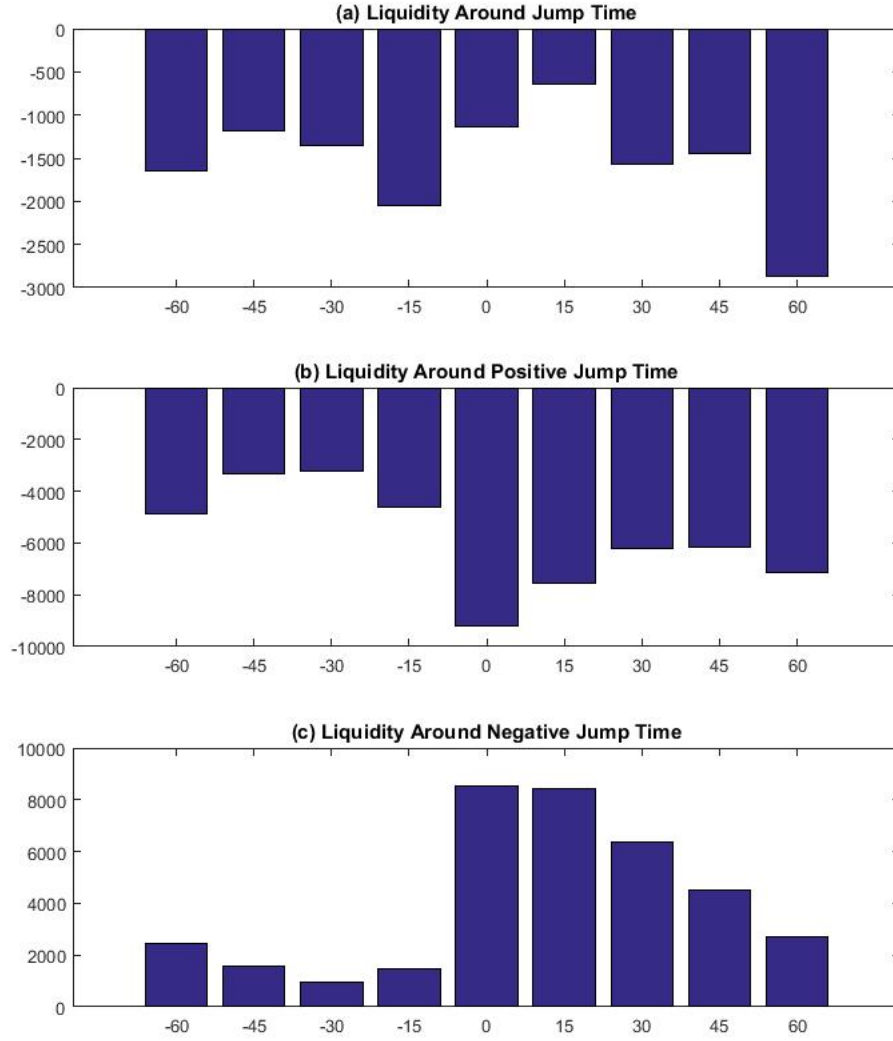


Figure 9: OB around the jump observations

The average order imbalances (OB) at 15, 30, 45 and 60 minutes before and after jump times are shown here.



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